

Section 1

MuTau Channel: proton mixing & weight
normalization issues

1.overview

Goal

Search for exclusive di-tau production in the muon-tau channel using 2018 data with the Precision Proton Spectrometer (PPS).

Samples Used

Sample	Description	Source
Data	2018 UL SingleMuon	Real collision data
DY	Drell-Yan Z/γ^* to tau-tau	MC simulation
ttbar	tt to tau-tau + X	MC simulation
QCD	Multi-jet background	Data-driven (same-sign)
Signal	Exclusive $\rightarrow$	MC simulation

2.Processing pipeline

NanoAOD
(CMS)

Phase 0
Skimming

Phase 1
Selection

Merge
Files

Plots

Proton
Mixing

Merged Phase 1 Files
(DY, ttbar, Data, QCD)

3. Phase0: initial skimming

Scripts

- fase0_data_mutau.py (Data)
- fase0_dy.py (Drell-Yan)
- fase0.py (ttbar)

Structure of scripts

- 1 **Trigger selection:** HLT_IsoMu24 (single muon trigger)
- 2 **Luminosity filtering:** Only certified good runs (dadosluminosidade.txt)
- 3 **Object selection:** At least 1 muon and 1 tau
- 4 **Variable extraction:**
 - Muon: pT, η , ϕ , mass, charge, ID
 - Tau: pT, η , ϕ , mass, charge, DeepTau IDs
 - Jets: pT, η , ϕ , mass, b-tag
 - MET: pT,

Proton variables saved (data only) $\rightarrow$ for MC samples these are added in with the proton mixing

4. phase1

scripts

- fase1_data.py (Data)
- fase1_dy.py (Drell-Yan)
- fase1_ttjets.py (ttbar)
- fase1_qcd.py (QCD)

Selection cuts applied

Variable	Cut	Description
muon_id	> 3	Medium MVA ID
tau_id1 (VSjet)	> 63	VeryTight DeepTau vs jets
tau_id2 (VSe)	> 7	Tight DeepTau vs electrons
tau_id3 (VSmu)	> 1	Loose DeepTau vs muons
muon_pt	$> 35 \text{ GeV}$	muon transverse momentum
tau_pt	$> 100 \text{ GeV}$	tau transverse momentum
muon_charge $\times$ tau_charge	< 0	opposite sign
$\Delta R(,)$	> 0.4	Angular separation

5.event weights in phase1

data

```
event_weight = 1.0 #Real data no correction needed
```

Drell-Yan(DY)

```
event_weight = generator_weight × mu_trig_sf × mu_idiso_sf × mu_reco_sf
```

Where: - generator_weight: MC generator weight (NLO corrections, etc.) -

mu_trig_sf: Muon trigger scale factor from POG - mu_idiso_sf: Muon ID and isolation scale factor - mu_reco_sf: Muon reconstruction scale factor

NOTE: Each SF is calculated with specific function present in the phase1 script

ttbar

```
event_weight = 0.15 × mu_trig_sf × mu_idiso_sf × mu_reco_sf
```

NOTE: Each SF is calculated with specific function present in the phase1 script (same structure as DY)

Important: The cross-section normalization factor (0.15) is **already included** at this stage → **this should be fixed, so the analysis steps are the same across samples, but it is not a priority now**

6.Proton Pool

- collection of real protons from 2018 PPS data, used to simulate pileup protons in MC.

Location

`/eos/cms/store/group/phys_smp/Exclusive_DiTau/proton_pool_2018/proton_pool_2018.root`

Contents

Branch	Description
proton_arm	detector arm
proton_xi	fractional momentum loss = $(p_{\text{beam}} - p_{\text{proton}})/p_{\text{beam}}$

How it was created (script also available in git repo- this code was kept as was)

- 1 Start with 2018 data with signal triggers

- Protons have distribution for 2018 running conditions
- includes detector acceptance and efficiency effects
- random sampling gives realistic pileup simulation

7. Proton Mixing Algorithm

- MC samples (DY, ttbar) don't have reliable pileup protons. they need to be added.

Script

```
merge_pp_mutau.py
```

Step 1: Always assign 1 proton per arm

```
#loops through proton pool until we find:  
# - 1 proton with arm=0 -> stored as xi_arm1_1  
# - 1 proton with arm=1 -> stored as xi_arm2_1
```

Step 2: Probabilistically add 2nd protons

variable name	probability	description
P	8.0%	1 proton on arm0, 1 proton on arm1
P	2.0%	1 proton on arm0, 2 protons on arm1
P	2.0%	2 protons on arm0, 1 proton on arm1

8.proton acceptance

- The probability that a random event has **at least 1 proton detected on both arms** simultaneously.

Why?

- after the proton mixing code, 100% of the mc events have 1 or more protons per arm
- this does not reflect number of pileup protons in real data samples
- we use a weight to account for the fraction of events that dont have pileup protons detected

Measurement from Data

Sample: Data_2018_UL_MuTau_nano_merged_proton_vars.root

Total events: 6,970

Events with 1 or more proton on both arms: 1,708

$$P = 1,708 / 6,970 = 0.245$$

$$P = 24.5\%$$

9. Bug in code -> pileup proton mixing

Original Code

```
# In merge_pp_mutau.py, line 143-144:  
weight = array("d", [float(FIXED_WEIGHT)])  
t_out.Branch("weight", weight, "weight/D")  
  
# In event loop:  
# weight was never updated -> was hard coded to 0.13
```

Problem: The original event_weight was discarded

Fixed code (corrected version)

```
#reads input event_weight if it exists  
has_event_weight = ensure_branch(t_in, "event_weight")  
if has_event_weight:  
    event_weight_in = array("d", [1.0])  
    t_in.SetBranchAddress("event_weight", event_weight_in)  
  
# In event loop:  
if event_weight_in is not None:
```

10. Where event weights are applied and values saved

data (no proton mixing needed)

Phase0: weight=1.0

Phase1: weight=1.0

Plotting: final_weight=1.0

Drell-Yan

Phase1: event_weight=generator_weight $\times$ muon_SF

Proton Mixing: weight=event_weight $\times$ 0.13

Plotting:

final_weight=weight $\times$ 1.81 (cross-section normalization)

ttbar

Phase1: event_weight=0.15 $\times$ muon_SF < -0.15 included in earlier step

Proton Mixing: weight = event_weight $\times$ 0.13

Plotting:

final_weight = weight $\times$ 1.0 -> no additional factor needed

11. Cross-Section normalization factors

formula used

$\text{weight} = N_{\text{MC}} / N_{\text{expected}}$

where $N_{\text{expected}} = \text{Luminosity} \times \text{Cross-section}$

values used

Sample	Factor	Included Where?
DY	1.81	Applied in plotting
ttbar	0.15	Already in Phase 1 event_weight

—> this (in principle) does not change the outcome but should be fixed later

12.status of plotting code

file: plot_m.py

how weights are currently used:

```
# QCD:fixed weight  
w_qcd=1.0  
# DY:reads weight from tree (currently contains only 0.13)  
w_dy = get_value(event, "weight", default=1.81)  
# ttbar:reads weight from tree (currently contains only 0.13)  
w_ttjets = get_value(event, "weight", default=0.15)
```

issues:

- 1 DY and ttbar files have `weight = 0.13` (not including `event_weight` -> explained earlier)
- 2 DY needs additional 1.81 factor not being applied
- 3 ttbar has 0.15 in `event_weight` but it's lost in current merged files

13.summary of issues found

Issue	Status	Action Required
proton mixing overwrites weights	code fixed	re-run proton mixing
DY cross-section factor missing	pending	apply 1.81 in plotting
proton acceptance value unclear	pending	decide: 0.13 or 0.245
ttbar event_weight lost	code fixed	re-run proton mixing

14.next steps

essential

- 1 decide on proton acceptance value (0.13 or 0.245)
- 2 **Re-run proton mixing** for DY and ttbar with fixed script
- 3 **Update plotting code** to apply correct normalizations

Verification

- 4 **generate comparison plots** before/after fix

15.Files

Processing scripts

Script	Purpose
fase0_data_mutau.py	Phase0 for Data
fase0_dy.py	Phase0 for DY
fase1_data.py	Phase1 for Data
fase1_dy.py	Phase1 for DY
fase1_ttjets.py	Phase1 for ttbar
fase1_qcd.py	Phase1 for QCD
merge_pp_mutau.py	Proton mixing
plot_m.py	Plotting

Output samples

File	Description
Data_2018_UL_MuTau_nanoData merged with real photons.root	
DY_2018_UL_MuTau_nanoData merged with real photons.root	

16.Proton acceptance measurement code

method in ROOT:

```
TFile* f = TFile::Open("Data_2018_UL_MuTau_nano_merged_proton_vars.root");
TTree* tree=(TTree*)f->Get("tree");
//count total events
Long64_t total = tree->GetEntries();

//count events with one or more proton on both arms
Long64_t good=tree->GetEntries(
    "Sum$(proton_multi_arm==0)>=1 && Sum$(proton_multi_arm==1)>=1"
);

//calculate acceptance
double P = (double)good / (double)total;
cout << "Proton acceptance P = " << P << endl;
```

result from my measurement:

```
total events:6,970
good events:1,708
```

- ① **Which proton acceptance value should be used?**
 - Matteo's 0.13 or measured 0.245?
- ② **Is the cross-section normalization correct?**
 - DY: 1.81
 - ttbar: 0.15 (already in event_weight)

